

Prove Equivalence between DFA and E-NFA

Theorem.

If $D = (Q_D, \Sigma, \delta_D, q_D, F_D)$ is the DFA constructed from E-NFA $E = (Q_E, \Sigma, \delta_E, q_E, F_E)$ by subset construction, then $L(D) = L(E)$.

Proof.

Actually, we have to prove

$$\hat{\delta}_D(q_D, w) = \hat{\delta}_E(q_E, w) \text{ for all } w \text{ where } |w| = n.$$

Proof by Structural Induction on $\hat{s}$

Basis step

Let $w = \epsilon$ where $|w| = 0$.

$$\text{L.H.S} = \hat{s}_D(q_D, \epsilon) = q_D \text{ (By basis definition of } \hat{s}_D)$$

$$= E\text{-Close}(q_E) \text{ (By subset construction).} \quad (1)$$

$$\text{RHS} = \hat{s}_E(q_E, \epsilon) = E\text{-Close}(q_E) \text{ (By basis definition of } \hat{s}_E) \quad (2)$$

$$(1) = (2)$$

$$\hat{s}_D(q_D, \epsilon) = \hat{s}_E(q_E, \epsilon).$$

Hence basis step is proved.

Inductive step.

Assume

$$\hat{s}_D(q_D, x) = \hat{s}_E(q_E, x) \text{ where } |x| = n \quad (3)$$

Let $w = xa$ where $|w| = n+1$ and $|x| = n$.

To prove,

$$\hat{s}_D(q_D, w) = \hat{s}_E(q_E, w) \text{ where } |w| = n+1$$

Proof

$$\begin{aligned} \text{By equation (3)} \Rightarrow \hat{s}_D(q_D, x) &= \hat{s}_E(q_E, x) \\ &= \{p_1, p_2, \dots, p_k\} \quad (4) \end{aligned}$$

$$\text{LHS} = \hat{\delta}_D(q_D, w)$$

$$= \delta_D(\hat{\delta}_D(q_D, x), a) \text{ By Inductive definition of } \hat{\delta}_D$$

$$= \delta_D(\{p_1, p_2, \dots, p_k\}, a) \text{ By eqn (4)}$$

$$= E\text{-Close}\left(\bigcup_{i=1}^k \delta_E(p_i, a)\right) \text{ By subset construction - (5)}$$

$$\text{RHS} = \hat{\delta}_E(q_E, w)$$

$$= E\text{-Close}(\delta_E(\hat{\delta}_E(q_E, x), a))$$

By inductive definition of $\hat{\delta}_E$

$$= E\text{-Close}(\delta_E(\{p_1, p_2, \dots, p_k\}, a)) \text{ By eqn (4)}$$

$$= E\text{-Close}\left(\bigcup_{i=1}^k \delta_E(p_i, a)\right) \text{ By rewriting - (6)}$$

From equation (5) and (6), LHS = RHS.
So, inductive step is also proved.

$$\hat{\delta}_D(q_D, w) = \hat{\delta}_E(q_E, w) \text{ for all } w \text{ where } |w| = n$$

Hence the Theorem